DATA ENGINEERING | BUSINESS INTELLIGENCE

Disaster Data Engineering & Intelligence Platform

Ten years of earthquakes and storms, from raw public data to a 4-page Power BI dashboard

1.7M
EARTHQUAKES

704K
STORM EVENTS

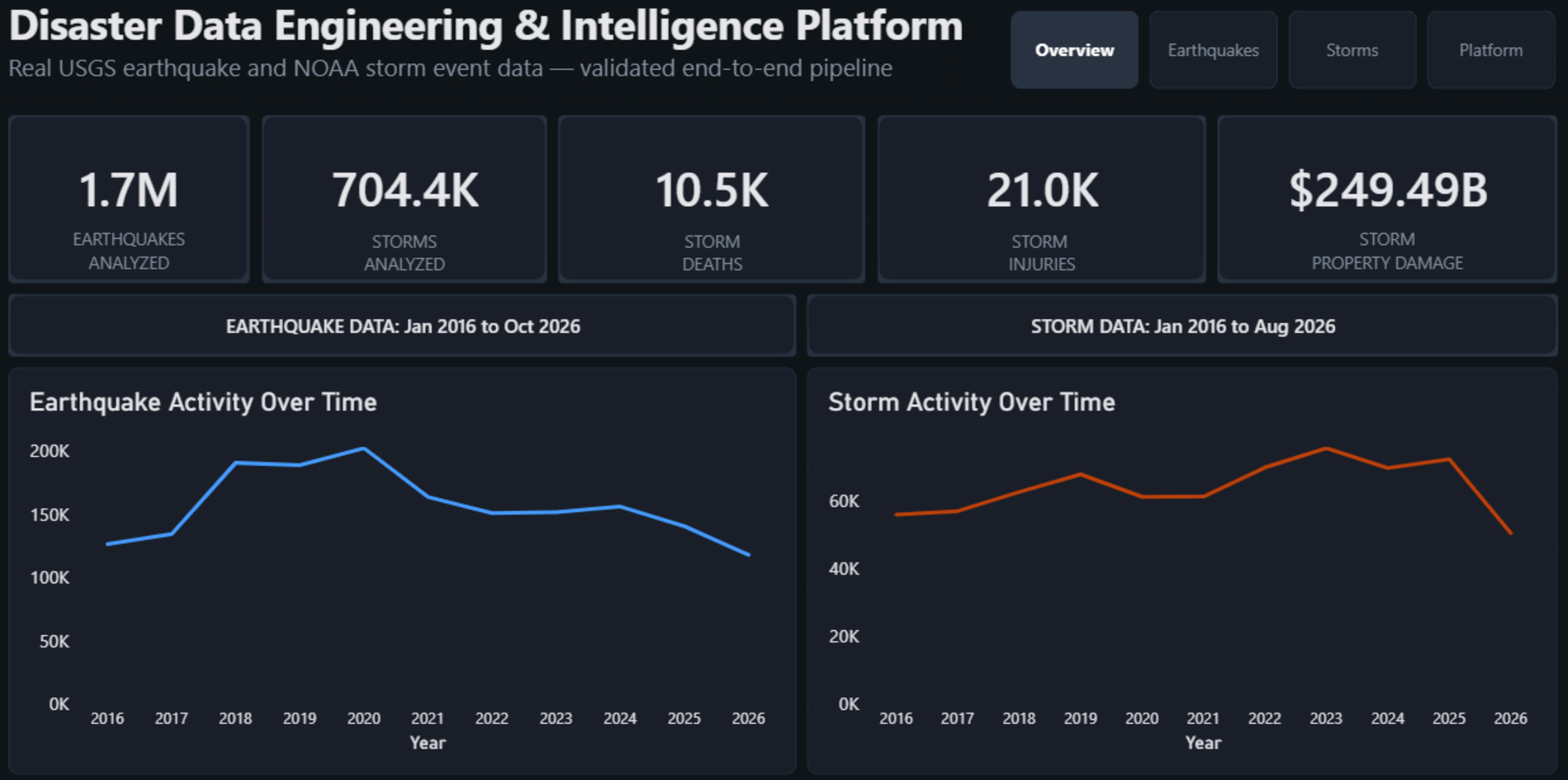
160
AUTOMATED TESTS

Python • PostgreSQL • SQL • Power BI • DAX • pytest

DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

One platform, two data sources

USGS earthquakes and NOAA storms, side by side from 2016 to 2026



Data through 31 Oct 2026 (earthquakes) and 1 Aug 2026 (storms). The latest year is year-to-date and updates with each pipeline run.

Note: earthquakes and storm events are analyzed independently. No direct event-level correlation is drawn between the two sources, as no defensible geographic or temporal join key exists between USGS and NOAA records.

DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

Alaska holds 31% of all recorded earthquakes

The strongest event was M8.8 near Kamchatka, Russia in 2025

Earthquake Analysis

Magnitude, depth and geographic characteristics of 1.7M real earthquakes, 2016 to 2026

Overview | **Earthquakes** | Storms | Platform

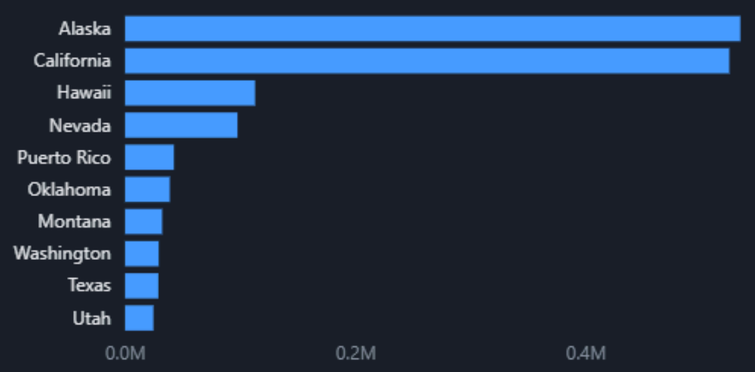
1,454
SIGNIFICANT EARTHQUAKES (M ≥ 6.0)

0.08%
% OF ALL EVENTS

1.63
AVERAGE MAGNITUDE

8.80
STRONGEST EVENT (MAGNITUDE)

Top 10 Regions by Earthquake Count



Year: All

Depth Distribution (km)



Event Locations



M6+ Events & Average Magnitude by Year



Where	Strongest	Peak Year
Alaska leads with 31% of all recorded events	Strongest event: M8.8, 2025 Kamchatka Peninsula, Russia Earthquake (2025)	Peak M6+ year: 2021 with 157 significant earthquakes

DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

Heat kills more people than tornadoes

Excessive Heat caused 2,247 deaths, the most of any storm type

Storm Analysis

Event types, seasonality and impact across 704K real storm events, 2016 to 2026

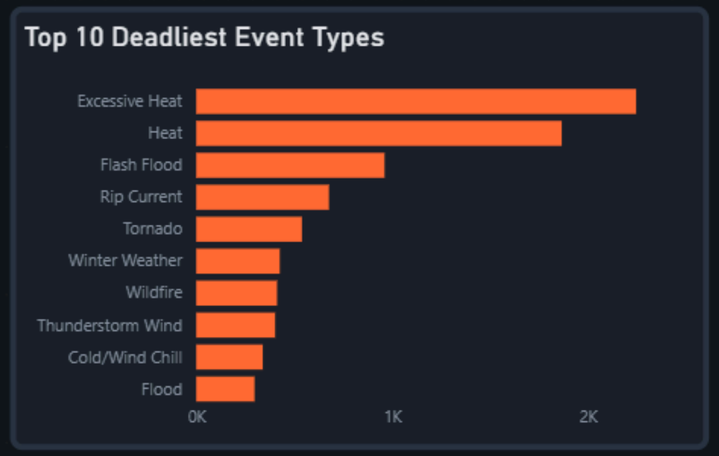
Overview | Earthquakes | **Storms** | Platform

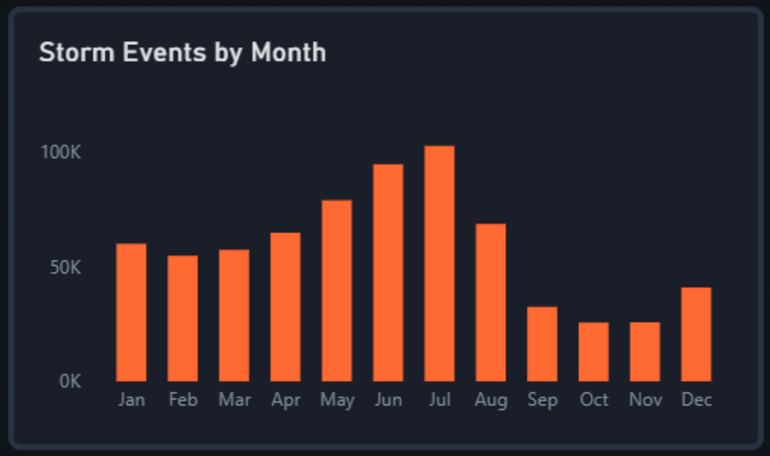
10.5K
TOTAL DEATHS

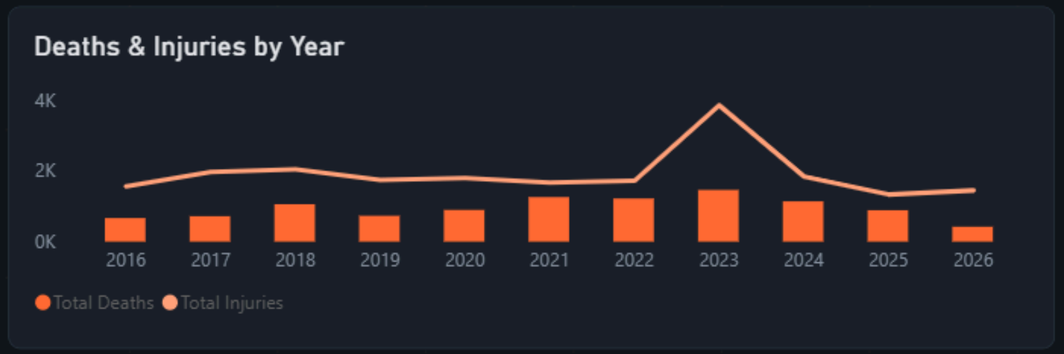
21.0K
TOTAL INJURIES

$249.49B
PROPERTY DAMAGE ($)

Thunderstorm Wind
MOST FREQUENT EVENT TYPE







Event Type: All

Month: All

Year: All

Deadliest	Injuries	Peak Season
Excessive Heat is the deadliest event type, with 2,247 deaths	Injuries peaked in 2023 with 3,857 recorded	Storm activity peaks in Jul (102,321 events across the period)

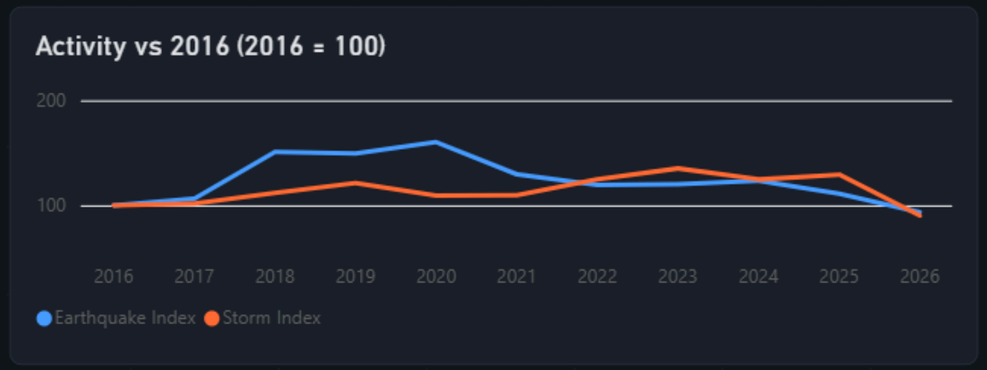
DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

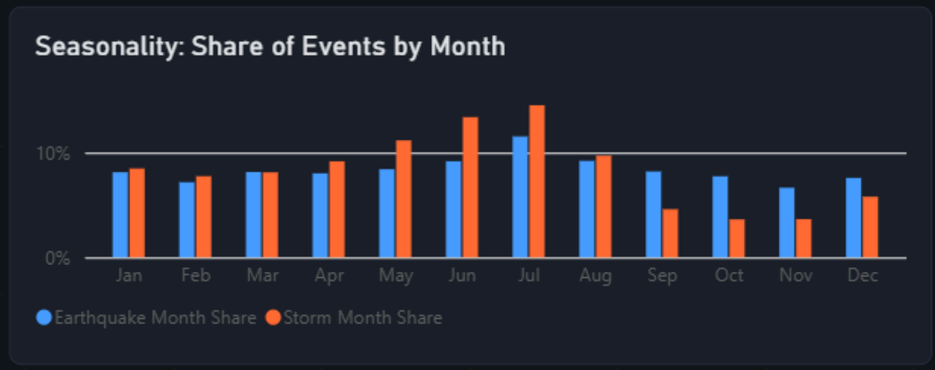
Earthquakes have no season. Storms do.

Earthquakes stay at 7-12% of events every month, while storms peak in July at 15%

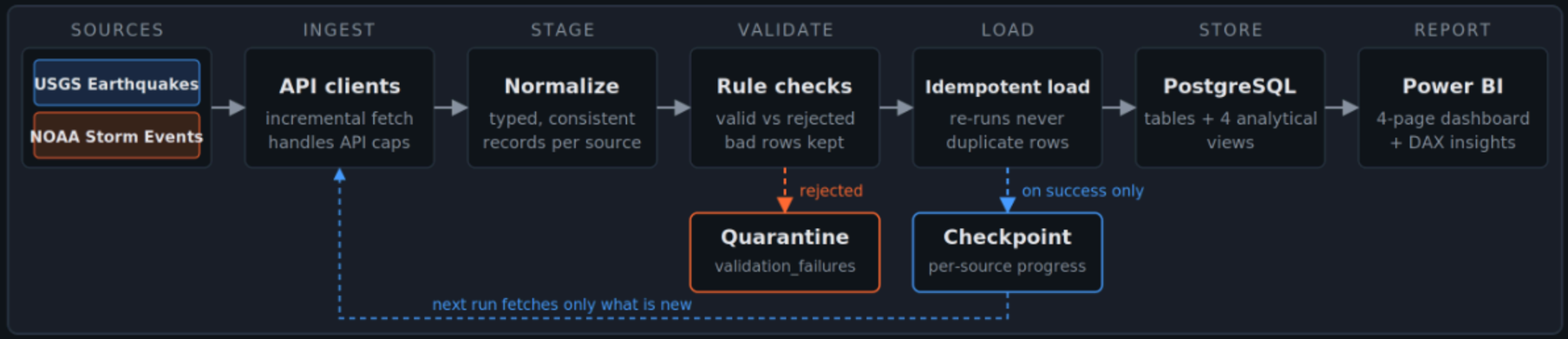
Platform Overview

Overview | Earthquakes | Storms | **Platform**





DATA PIPELINE



Earthquakes	Storms	Seasonality
Earthquake activity peaked in 2020 (+60% vs 2016)	Storm activity peaked in 2023 (+35% vs 2016)	Storms peak in Jul (15% of events); earthquakes stay between 7% and 12% every month

DATA PIPELINE

Built to fail safely

Incremental checkpoints, idempotent loads, a quarantine for bad records, and failure isolation between sources



CHECKPOINTS

Resume safely after successful progress

IDEMPOTENCY

Repeat runs without duplicating loaded data

QUARANTINE

Bad records are isolated instead of silently loaded

Python • PostgreSQL • SQL • Power BI • DAX • pytest

github.com/Sravani-2205/disaster-data-engineering-platform